

The Price of Credit: Interest Rate Dispersion Across Formal and Informal Lenders in India

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Abstract

Indian households pay substantially higher interest rates on informal loans than on formal loans. We use loan-level data from the 2019 All-India Debt and Investment Survey to measure this gap and examine how it varies across borrowers, loan purposes, and locations. The average annual interest rate is 20.08 percent for informal credit and 11.78 percent for formal credit, a gap of 8.29 percentage points, or about 70 percent of the formal rate. Controlling for loan terms, household characteristics, and district fixed effects yields an informal premium of 15.7 percentage points. The premium ranges from 13.7 to 20.0 percentage points across loan purposes and is largest for medical treatment. Collateral is associated with lower rates in both markets, but the estimated discount does not differ significantly by lender type. Rural borrowers face a larger informal premium within the same state. Moreover, 9.8 percent of households hold both formal and informal debt, showing that access to formal credit can coexist with reliance on informal borrowing. These findings point to loan purpose and rural credit markets as central considerations in efforts to reduce household borrowing costs.

Keywords: informal credit, interest rate gap, financial inclusion, rural credit markets, India, AIDIS

* The views expressed here are solely the responsibility of the author and should not be interpreted as reflecting the views of the Center for Advanced Financial Research and Learning (CAFRAL) or the Reserve Bank of India.

Introduction

Access to affordable credit remains one of the central constraints on household welfare and productive investment in rural and low-income India. Despite decades of financial-inclusion policy nationalised banking, priority-sector lending mandates, regional rural banks, self-help-group linkage programmes, and more recently digital and branchless banking a substantial share of Indian households continues to borrow from informal sources: moneylenders, landlords, traders, and kin. These informal channels typically operate outside prudential regulation and, where they charge a positive price for credit at all, often do so at rates that are large multiples of formal-sector rates.

Understanding the size, distribution, and correlates of the resulting formal-informal interest rate gap is therefore central to evaluating whether India's financial-inclusion effort is closing, or merely papering over, a persistent two-tier credit market. This matters because a wide gap left unexamined is easy to misread: a raw difference in rates paid by two groups could reflect unequal pricing once both groups reach both markets, or it could instead reflect unequal selection into the cheaper market in the first place, and these two explanations point to very different policy remedies. Motivated by this ambiguity, and by the fact that the most recent nationally representative loan-level data on this question the 2019 AIDIS has not yet been used to separate these mechanisms in a single, loan-level regression framework, this paper sets out to characterise the gap in detail.

This paper uses loan-level microdata from the 2019 All-India Debt and Investment Survey (AIDIS), the primary official source on household indebtedness in India, to characterise this gap in detail. Three questions organise the analysis. First, how large is the formal–informal interest rate gap in the aggregate, and how does it vary across rural and urban areas, lender types, social groups, loan purposes, and states. Second, which households end up borrowing from both markets simultaneously, and what does their demographic profile reveal about the forces that push a household toward dual-market access. Third, conditional on being exposed to both markets, which household and state characteristics predict a household's own interest rate gap and does this differ from what predicts raw, unconditional differences in rates across groups.

Distinguishing these questions matters for policy. A wide raw gap for a particular group could reflect unequal treatment or worse terms once that group participates in both credit markets, or it could instead reflect selection differences in who gains access to formal credit in the first place, leaving informal credit as a residual for those who are excluded. The two mechanisms call for different remedies. The first points toward regulating or subsidising informal lenders or improving bargaining power for disadvantaged borrowers within the informal market, while the second points toward expanding formal-sector reach and easing the collateral and documentation barriers that keep certain groups out of the cheaper channel altogether. This paper's loan-level regression design – a main regression with a portfolio-composition extension, a second regression on collateral, a third regression on purpose heterogeneity, and complementary district- and state-level aggregations – is built specifically to separate these two stories, whether the observed gap reflects unequal pricing conditional on loan and household

characteristics (the portfolio-composition, collateral, and purpose-heterogeneity results in Tables 13, 14, and 15) or a more diffuse, geographically concentrated access channel (the district- and state-level evidence in Tables 16 and 17).

The remainder of this section reviews the theoretical and empirical literature on informal credit markets and the sources of the formal-informal price gap. Section 2 describes the AIDIS data, sample construction, and empirical strategy, including the loan-level regression framework and its extensions used later in the paper. Section 3 presents descriptive and regression results. Section 4 discusses the findings and their implications. Section 5 concludes and outlines directions for further analysis, including a set of additional regressions that have not yet been estimated but would extend the analysis.

The persistence of high-priced informal lending alongside a regulated, lower-priced formal banking sector has long puzzled development economists, since a competitive credit market should, in principle, arbitrage away such a gap. The dominant explanation in the literature is informational: formal lenders operate at arm's length from borrowers and must rely on collateral, documentation, and credit history to screen risk, which is costly to produce and verify for small, non-salaried, or rural borrowers (Hoff & Stiglitz, 1990). Informal lenders, by contrast, typically live or work alongside their borrowers, allowing them to observe income shocks, repayment capacity, and effort directly and to enforce repayment through repeated, personalised dealing and social sanction (Aleem, 1990). The interest premium charged informally is, in this view, substantially a payment for local information and enforcement rather than pure monopoly rent, although both forces plausibly coexist in markets where a small number of moneylenders serve a given village (Bell, 1990).

A related literature emphasises interlinkage: informal lenders particularly landlords and traders often bundle credit with tenancy, output marketing, or input-supply relationships, allowing them to recover an implicit return through channels other than the stated interest rate (Basu, 1997; Swaminathan, 1991). This helps explain why some informal categories, notably lending among relatives and friends, are reported at or near a zero interest rate: such lending is sustained by reciprocity and social obligation rather than a price mechanism, and should not be pooled together with commercial moneylender or landlord credit when characterising the informal rate.

A second strand of literature traces the gap to the uneven geographic expansion of formal banking. Burgess and Pande (2005) show that India's post-1977 social/rural banking programme, which mandated the opening of a specified number of unbanked-location branches for each urban branch a bank opened, reduced rural poverty in states that received relatively more rural branches, and that the effects of this branch expansion persisted long after the mandate was reversed in the early 1990s. States and districts that gained proportionally fewer rural branches during this period continued to exhibit weaker formal-credit penetration decades later, plausibly leaving a larger share of the marginal borrowing population dependent on informal sources and widening the observed rate gap in those regions.

More recent work makes the mechanism explicit rather than inferred: using a regression-discontinuity design built around India's post-2005 bank-licensing rules, Briones, Chiseliov,

Gushchin, Guttesen, and Narayanaswamy (2024) find that a new branch opening lowers informal lending rates in the surrounding area by roughly 24% within five years, and attribute most of this fall to competition between banks and moneylenders rather than to any direct change in formal rates. That is essentially the supply-side channel the district-level regression in Section 3.9 is getting at, even though a direct branch-density measure is not yet in the data used here (see Section 5).

A distinct body of work documents that social identity, and caste in particular, continues to shape access to and the terms of formal credit in India net of measurable economic characteristics such as income, landholding, or occupation (Deshpande, 2011; Kumar, 2016). Where formal-credit allocation depends on soft information, references, or bank-manager discretion, historically marginalised groups may be disproportionately screened out of the cheaper channel and pushed to compete for the same informal supply as everyone else.

Two recent papers sharpen this picture considerably. Working with the same 77th-round NSS data used in this paper, Zeeshan, Khan, Shamim, and Khatoon (2025) apply a Blinder-Oaxaca decomposition to institutional credit access and find that an unexplained, potentially discriminatory component accounts for 42% of the SC-versus-Other gap and 60% of the ST-versus-Other gap, against only 8% for OBC households an ordering that lines up closely with both the raw gaps in Table 4 and the informal-penalty results in Section 3.9. Sangwan and Saha (2026), looking directly at formal-bank and informal-moneylender loan applications, report a more surprising result: in their data, formal banks discriminate against lower-caste applicants more than moneylenders do. That finding complicates any easy assumption that informal lenders are simply the more exploitative side of the market, and is worth keeping in mind alongside this paper's own finding that it is specifically the informal penalty, not the formal-sector rate on its own, that rises most sharply for SC and ST borrowers (Section 3.9). This literature motivates treating social group as a candidate explanatory variable for the gap, while also raising the empirical question addressed directly in this paper of whether caste operates mainly through unequal access to formal credit (a selection channel) or through unequal terms conditional on already borrowing informally and formally (a pricing channel).

A separate literature, associated with the "portfolios of the poor" framework, argues that low- and middle-income households routinely hold multiple, imperfectly substitutable financial instruments simultaneously in order to manage irregular income and lumpy expenditure needs, rather than relying on a single credit source (Rutherford, 2000; Karlan & Morduch, 2010). Under this view, formal and informal credit can function as complements rather than pure substitutes: formal credit serving larger, longer-horizon, or collateralisable needs, and informal credit providing rapid, flexible top-up financing. This framework motivates treating the subpopulation of households that hold both formal and informal debt as a distinct group worth profiling on its own terms, rather than folding it into either the formal-only or informal-only category, and is directly reflected in this paper's household portfolio classification and household-level regression sample.

A policy-oriented literature further documents that Indian agricultural credit has long been directed and subsidised through priority-sector lending targets and interest subvention schemes aimed specifically at farm capital and revenue needs (Shah, Rao, & Vijay Shankar, 2007).

Because these schemes compress the formal-sector rate for agricultural purposes specifically, they can mechanically widen the measured formal–informal gap for agricultural loans even absent any change in the informal rate, a consideration that bears directly on how the loan-purpose results in this paper should be interpreted.

Finally, and complementing the interlinkage and information-based explanations above, a theoretical literature on joint-liability lending shows how group-based contracts the model underlying much of the microfinance and self-help-group (SHG) movement in India can substitute for collateral by making borrowers mutually responsible for each other's repayment, thereby using peer information and social capital to support lower interest rates than an individually collateralised formal loan would otherwise require (Ghatak & Guinnane, 1999). This literature is relevant to interpreting the relatively low rates observed for SHG/JLG lending in the present data (Table 3) and to the broader question of whether formal-sector product design, rather than formal-sector reach alone, can narrow the gap (Banerjee & Duflo, 2010).

The clearest causal evidence on this point comes from a randomised rollout of a government SHG programme in Bihar: Hoffmann, Rao, Surendra, and Datta (2021) find that two years after the programme began, informal borrowing fell and fewer informal lenders remained active in treatment villages, and that the average interest rate on new loans was about 8 percentage points lower than in control areas with SC/ST households, who had faced the highest informal rates at baseline, seeing the largest declines. It is among the clearest demonstrations that expanding a specific formal-sector product, rather than formal-sector reach in general, can close the gap fastest for exactly the households this paper finds face the steepest informal penalty.

2. Data and Empirical Strategy

This section presents descriptive and regression evidence on the gap between formal and informal interest rates, drawn from the 2019 All-India Debt and Investment Survey (AIDIS, NSS 77th Round). We first document the size of the gap in the aggregate and across rural/urban areas, lender types, social groups, loan purposes, and states. We then move to the household as the unit of analysis, describing how households sort into formal-only, informal-only, dual-access ("Both"), and non-borrowing categories, and profile the dual-access group demographically, before presenting regressions of the gap on state- and household-level covariates.

2.1 Data and Sample Construction

The analysis draws on loan-level records from AIDIS 2019. Each loan is classified as formal (credit agency codes 1-8 and 10-13: scheduled commercial banks, regional rural banks, cooperative societies and banks, insurance companies, provident funds, employers, financial corporations/institutions, NBFCs/MFIs, and both bank-linked and non-bank-linked SHG/JLG groups) or informal (code 9 and codes 14-20: other non-institutional agencies, landlords, agricultural and professional moneylenders, input suppliers, relatives/friends, chit funds, and market commission agents/traders). Code 9 ("other, non-institutional") is non-institutional by definition and is treated as informal throughout, not grouped with the formal block. Sections

3.1-3.8 below use the full set of 105,844 loan records with a non-missing, non-zero reported interest rate, winsorised at the 1st and 99th percentiles to limit the influence of extreme or likely miscoded values; the core outcome of interest in that descriptive analysis is the interest rate gap, defined both as a percentage-point gap (informal mean rate minus formal mean rate) and as a percentage gap relative to the formal rate.

Section 3.9's loan-level regressions apply a further restriction: loans must be reported as still outstanding on the survey's 30 June 2018 reference date, so that the reported rate and loan terms are contemporaneous. This narrows the loan universe to 72,040 records (43,191 formal, 28,849 informal); of these, 58,949 carry a non-zero, non-missing interest rate, and 13,090 are treated as interest-free either because the household reported the loan that way, or because the reported rate is exactly zero, which we treat the same way. The regressions further require non-missing values on every household control, and land ownership turns out to be the binding constraint: it is missing for 6,855 of the 58,949 rated loans, and the missingness is not random. Loans dropped for this reason are disproportionately urban (73% of dropped loans, against 34% of the loans that are kept) and somewhat more likely to be informal (33% of dropped loans versus 28% of kept loans), so the estimation sample used in Section 3.9 skews slightly more rural and slightly more formal than the full rated-loan sample. We do not think this changes the headline estimates much the robustness checks at the end of Section 3.9, including a missing-indicator specification that keeps all 58,949 rated loans, move the informal coefficient by well under one percentage point but it is a genuine limitation of the AIDIS land-ownership item worth flagging rather than glossing over. After this restriction, the estimation sample used in most of Section 3.9's specifications is 52,092-52,094 loan-records; the small variation across specifications (52,093 rather than 52,094 in several tables) comes from one district being dropped as a singleton fixed effect whenever district fixed effects are included.

Six loan-level and aggregate regression specifications are used to identify correlates of the gap, corresponding to Tables 12-17, using the classification and estimation samples described above. The main regression and its extensions are estimated on the 52,092-52,094 loan-records with complete covariates; the aggregation exercises in Section 2.3 work from the same underlying loans, recombined at the district or state level.

2.2 Loan-Level Specification

The baseline specification regresses the loan-level annual interest rate on an indicator for whether the lender is informal, loan controls, household controls, and district fixed effects:

$$r_{ij} = \beta_0 + \beta_1 \text{Informal}_{ij} + \beta_2 L_{ij} + \beta_3 X_i + \delta_d + \varepsilon_{ij} \quad (1)$$

where r_{ij} is the annual interest rate (%) on loan j held by household i , winsorised at the 1st and 99th percentile among interest-bearing loans; Informal_{ij} is an indicator equal to one for non-institutional lenders; L_{ij} is a vector of loan controls (purpose, tenure, whether secured, and the log amount borrowed); X_i is a vector of household controls (household size, log monthly per-capita consumption expenditure, maximum years of education among household members, ST/SC/OBC social-group indicators, and land owned); and δ_d is a district fixed effect. Standard errors are clustered on district (680 clusters). β_1 is the headline formal-informal rate gap, net of

loan characteristics, household characteristics, and any time-invariant district-level factor such as bank density or moneylender competition.

The main regression extends the baseline specification with a loan-portfolio-composition specification (Table 13), adding the household's institutional-loan share, debt-to-asset ratio, number of loans held, and a prior-loan-write-off indicator to test whether the gap reflects a risk-premium or default-history channel operating through the household's broader borrowing profile, on top of the loan and household controls in equation (1). Two further specifications probe additional channels. A second regression, the collateral specification (Table 14), interacts Informal_{ij} with an indicator for whether the loan is secured, to test whether informal lenders discount collateral differently from formal ones. A third regression, the purpose-heterogeneity specification (Table 15), fully interacts Informal_{ij} with a set of loan-purpose dummies, recovering a separate informal-gap estimate for each of twelve purposes.

2.3 District- and State-level Aggregation

Two further exercises aggregate the loan-level result to describe geographic structure in the gap. The first is a two-step district exercise (Table 16). In the first step, a leaner version of equation (1) omitting loan controls, so that the district fixed effects are not contaminated by within-district variation in loan terms is estimated with district fixed effects absorbed; the recovered coefficients δ^d (680 districts) are interpreted as each district's residual "credit cost" net of household composition. In the second step, δ^d is regressed on district-average characteristics:

$$\delta^d = \pi_0 + \pi_1 \text{MPCE}^d + \pi_2 \text{Land}^d + \pi_3 \text{ShareST/SC}^d + \pi_4 \text{Rural}^d + u_d \quad (2)$$

where MPCE^d and Land^d are district-average monthly per-capita consumption and land owned, ShareST/SC^d is the district's ST/SC population share, and Rural^d is the district's rural population share. A companion specification (D-1, not yet estimated see Section 5) will add bank branch density per lakh population from external RBI/DBIE data once merged, to test directly whether formal banking supply compresses the district-level gap; a further district specification (D-3), regressing the district's institutional loan-value share on branch density, SHG coverage, and district demographics, awaits the same external file.

The second exercise (Table 17) re-estimates equation (1) with state fixed effects in place of district fixed effects and adds an interaction between Informal_{ij} and a rural/urban sector indicator, Rural_i , to test whether the informal penalty is steeper in rural areas even net of state-level regulatory and banking-structure differences:

$$r_{ij} = \beta_0 + \beta_1 \text{Informal}_{ij} + \beta_2 \text{Rural}_i + \beta_3 (\text{Informal}_{ij} \times \text{Rural}_i) + \beta_4 X_i + \mu_s + \varepsilon_{ij} \quad (3)$$

with standard errors clustered on state (36 clusters). A complementary state cross-section on the average gap (S-1), which would add state poverty rates and priority-sector lending intensity from external data, has not yet been estimated (see Section 5).

3. Results

3.1 Sample Overview

The analysis sample contains 105,844 individual loan records reported by 116,461 sample households, of which 72,821 households (62.5%) reported at least one outstanding loan as of the survey reference date. Weighting loans by their survey multiplier, 63.9% of loans (by count) were formal and 36.1% informal (Table 1). Informal borrowing is markedly more prevalent in rural areas: 38.4% of rural loans are informal, compared with 28.9% of urban loans.

Table 1. Sample summary: formal vs. informal loans, weighted

Area	Total loans	Informal (%)	Formal (%)
All-India	1.78×10^{10}	36.1	63.9
Urban	4.42×10^9	28.9	71.1
Rural	1.34×10^{10}	38.4	61.6

Notes: Loan counts are weighted by the AIDIS household multiplier (MLT); percentages are shares of weighted loan counts within each row.

3.2 The Formal–Informal Interest Rate Gap

Table 2 reports the weighted mean and median annual interest rate paid on formal and informal loans. Informal credit carries an average rate of 20.08% per annum, compared with 11.78% for formal credit a gap of 8.29 percentage points, or informal rates being 70.4% higher than formal rates. The gap is wider in rural areas (8.97 pp, 77.5%) than in urban areas (5.82 pp, 47.3%), consistent with thinner formal-sector penetration and greater reliance on moneylenders and other informal sources outside cities.

Table 2. Mean and median interest rates, formal vs. informal (% p.a.)

Area	Informal mean	Informal median	Formal mean	Formal median	Gap (pp)	Gap (%)
All-India	20.08	24.00	11.78	11.00	8.29	70.37
Rural	20.56	24.00	11.59	11.00	8.97	77.45
Urban	18.13	18.00	12.30	12.00	5.82	47.32

Notes: Means/SDs are weighted by MLT; medians via weighted percentiles. Gap (pp) = Informal mean – Formal mean; Gap (%) = Gap (pp) / Formal mean $\times$ 100.

This gap is best understood as the price of information rather than of capital alone. Formal lenders screen borrowers using collateral and documentation, and hesitate to lend where these are scarce or costly to verify; informal lenders, by contrast, typically live and work alongside their borrowers and can observe repayment capacity and enforce repayment through repeated, personalised dealing (Aleem, 1990; Hoff & Stiglitz, 1990). The premium a borrower pays informally is therefore, in part, a payment for this local information and enforcement advantage rather than simple monopoly rent, though both forces plausibly operate together (Bell, 1990). The rural–urban difference in Table 2 is consistent with this reading: rural India has historically had thinner formal bank presence per capita, so the marginal borrower who cannot obtain

formal credit is pushed further out along the informal supply curve, widening the observed gap (Burgess & Pande, 2005).

3.3 Interest Rates by Lender Type

Disaggregating by the specific lending institution or agent (Table 3) shows that the aggregate formal–informal gap masks substantial heterogeneity within each segment. Among formal lenders, Banks (9.85%), Regional Rural Banks (8.58%), and Cooperatives (8.53%) charge the lowest rates, while "Other Formal" institutions (18.44%) and NBFCs/MFIs (17.06%) are considerably more expensive — rates that in fact exceed several informal categories. Among informal lenders, professional and agricultural moneylenders charge by far the highest average rate (32.22%), followed by landlords (27.41%) and traders/agents (23.00%). Loans from relatives and friends, by contrast, are reported at a 0.00% average rate, consistent with the largely non-commercial, kinship-based nature of this lending channel; this category alone accounts for 18,546 loan records, the single largest informal source by volume.

Table 3. Mean interest rate by lender type

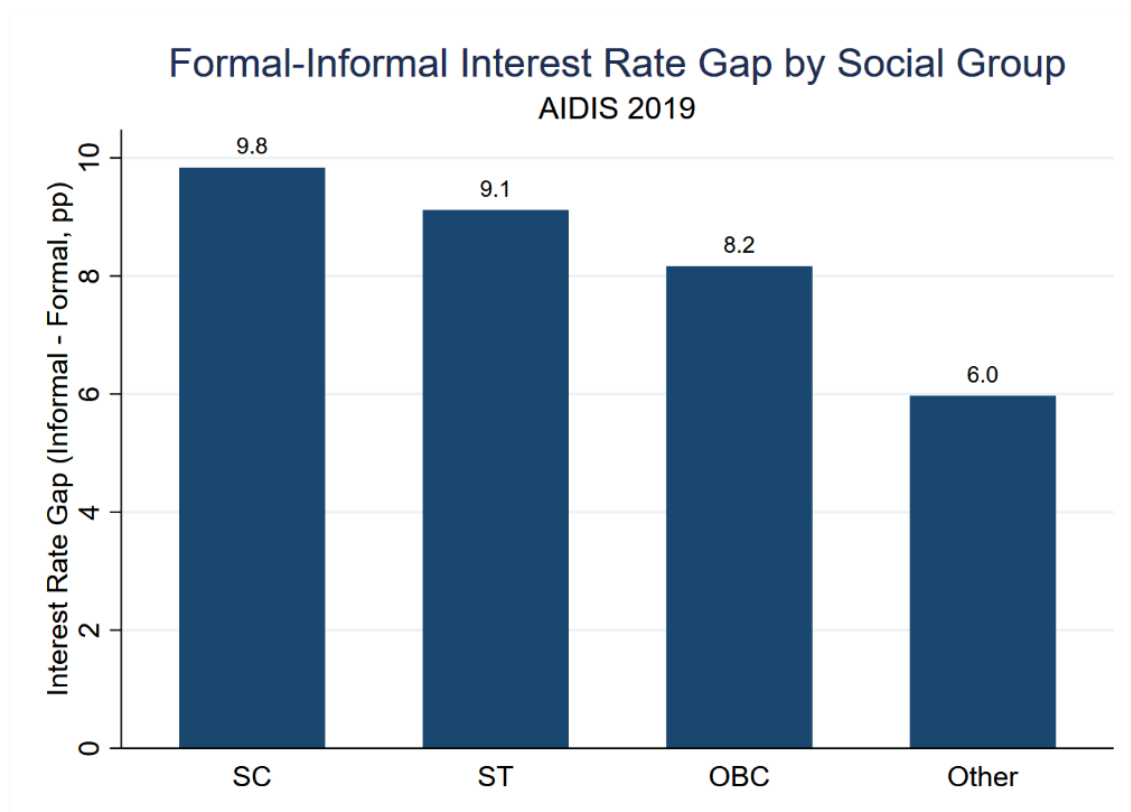
Lender type	Mean rate (%)	SD	N
Bank	9.85	4.15	31,359
RRB	8.58	4.04	3,957
Cooperative	8.53	5.44	7,748
Other Formal	18.44	12.70	7,619
Employer	9.42	7.39	145
NBFC/MFI	17.06	7.50	5,371
SHG/JLG	13.55	6.97	9,401
Landlord	27.41	18.69	1,081
Moneylender	32.22	20.27	18,815
Input Supplier	23.82	15.11	669
Relatives/Friends	0.00	0.00	18,546
Chit Fund	18.59	12.79	374
Trader/Agent	23.00	12.94	758

Notes: N is the unweighted number of loan records; means/SDs are weighted by MLT.

That relatives-and-friends lending clears at a zero rate while moneylenders and landlords charge 27–32% illustrates why "informal credit" should not be treated as a single market: kinship-based lending is sustained by reciprocity and social sanction rather than a price mechanism, whereas moneylender and landlord credit is frequently interlinked with tenancy, output marketing, or labour contracts that let the lender recover an implicit return even when the stated interest rate looks unremunerative on its own (Basu, 1997; Swaminathan, 1991). Failing to separate these categories, as a single "informal" average would, understates just how costly credit is for households without access to family networks or interlinked arrangements.

3.4 The Gap Across Social Groups

Table 4 shows the gap by the household head's social group. Scheduled Caste (SC) households face both the highest informal rate (23.50%) and the widest absolute gap (9.83 pp, 71.96% of the formal rate), followed closely by Scheduled Tribe (ST) households (9.11 pp, 75.80%). OBC households face a somewhat smaller gap (8.16 pp), and "Other" (general-category) households the narrowest (5.97 pp, 54.23%). This ordering is consistent with SC/ST households facing greater constraints in accessing formal credit and being pushed toward more expensive informal sources.



This pattern is consistent with a wider literature documenting that caste continues to shape access to, and terms of, formal credit in India, independent of measurable economic characteristics such as income or landholding (Deshpande, 2011). Where formal credit allocation depends on soft information, references, or local bank-manager discretion, historically excluded groups are disproportionately screened out of the cheaper channel and left to compete for the same informal supply as everyone else—a pattern this paper investigates further through household credit-portfolio composition (Section 3.7–3.8, Tables 7–11) and through the loan-level regressions in Section 3.9.

3.5 The Gap by Loan Purpose

The gap also varies sharply by the stated purpose of the loan (Table 5). Loans taken for agricultural purposes show the widest gaps: Farm Revenue loans carry a gap of 14.55 pp (176.9%), and Farm Capital loans a gap of 11.93 pp (127.9%) both far above the sample-wide average. Litigation (10.19 pp) and Non-farm Revenue (10.97 pp) loans are similarly costly

relative to their formal-sector counterparts. At the other end, Medical (6.02 pp, 39.7%) and Household Expenditure (5.76 pp, 41.9%) loans show comparatively narrow gaps, in part because formal-sector rates for these purposes are themselves higher than for productive/agricultural lending.

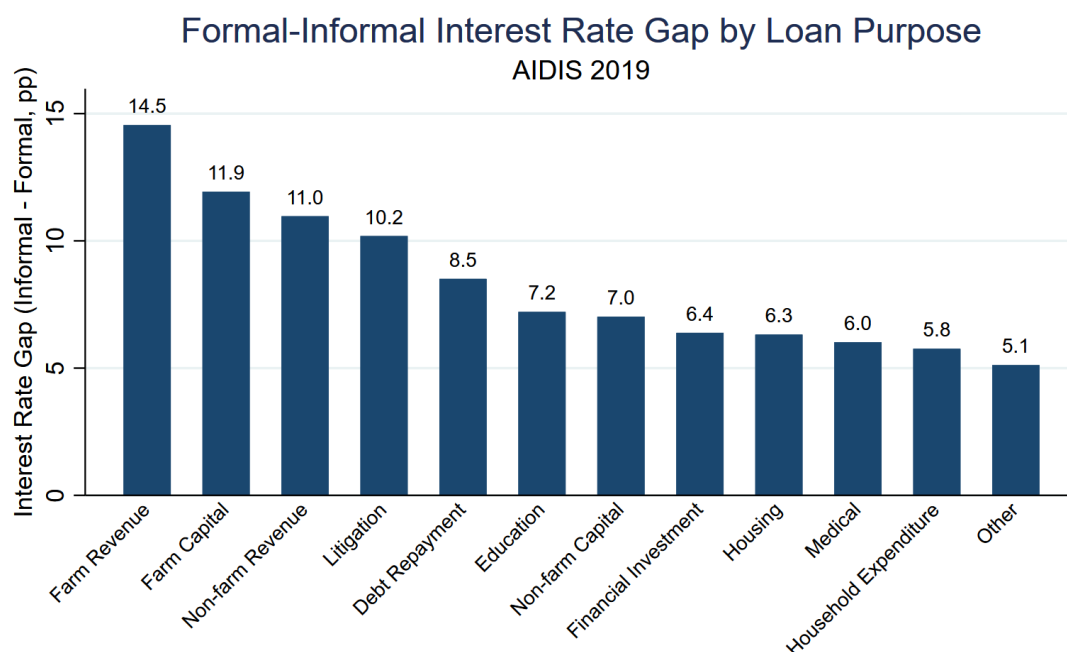


Figure 2. Formal–informal interest rate gap by loan purpose

Agricultural credit in India has long been subsidised and directed through priority-sector lending targets aimed precisely at farm revenue and capital needs (Shah, Rao, & Vijay Shankar, 2007), which mechanically compresses the formal rate for these purposes and, holding the informal rate fixed, mechanically widens the gap in Table 5. The width of the agricultural gap should therefore be read partly as a policy artefact of directed, subsidised formal agricultural credit rather than solely as evidence of an unusually predatory informal market for farm loans — though, as Table 13 shows, the agricultural share of a household's borrowing remains the single strongest predictor of its gap even after other covariates are held fixed, suggesting a real, not purely mechanical, effect as well.

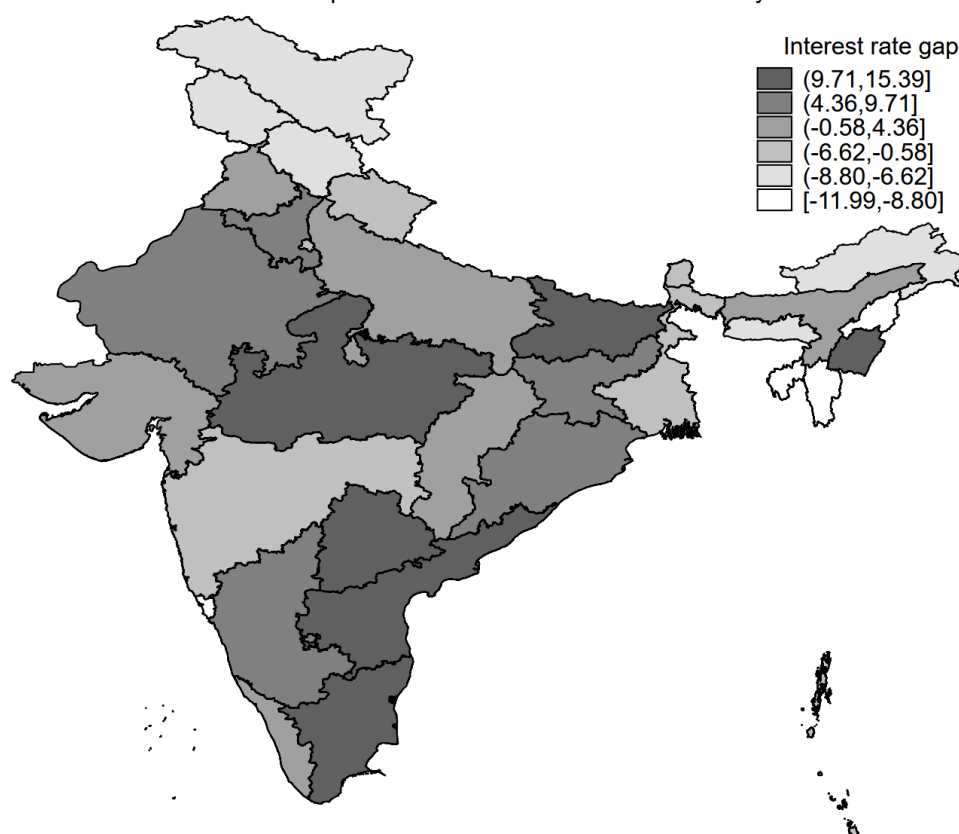
3.6 State-wise Variation in the Gap

There is substantial geographic heterogeneity in the gap (Table 6). Andhra Pradesh has both the highest informal rate (26.25%) and the largest gap (15.39 pp, 141.7%), followed by Bihar (14.77 pp, 92.4%), Madhya Pradesh (14.52 pp, 133.5%), and Telangana (12.93 pp, 119.7%) all large, agriculturally-oriented states where informal moneylending appears deeply entrenched. At the opposite extreme, several smaller states and Union Territories (Lakshadweep, Goa, Nagaland, Chandigarh) show a negative gap, i.e., formal rates exceed informal rates; in these cases informal borrowing is both rare and dominated by low/zero-interest relatives-and-friends lending, so these estimates are noisy and are flagged separately as "thin-sample" states in the regression analysis below.

Table 6. State-wise interest rate gap (top and bottom five states)

Rank	State	Formal (%)	Informal (%)	Gap (pp)	Gap (%)
1	Andhra Pradesh	10.86	26.25	15.39	141.68
2	Bihar	15.99	30.76	14.77	92.41
3	Madhya Pradesh	10.88	25.40	14.52	133.46
4	Telangana	10.80	23.73	12.93	119.68
5	Manipur	20.28	30.39	10.11	49.85
32	Dadra & Nagar Haveli	12.56	3.27	-9.29	-73.94
33	Chandigarh	11.41	1.72	-9.69	-84.88
34	Nagaland	12.19	1.11	-11.08	-90.89
35	Goa	11.71	0.08	-11.63	-99.32
36	Lakshadweep	11.99	0.00	-11.99	-100.00

Interest Rate Gap between Informal and Formal Credit by State



Source: AIDIS 2019 (NSS 77th Round). Gap = Mean informal interest rate – Mean formal interest rate.

Figure 3. Choropleth map of the state-wise formal–informal interest rate gap

The geography in Figure 3 lines up closely with India's historical "bank-branch expansion" experience: districts and states that gained proportionally fewer rural branches during the post-1977 social-banking period continued to show weaker formal-credit penetration and worse rural outcomes decades later (Burgess & Pande, 2005), and the states with the widest gaps here — Andhra Pradesh, Bihar, Madhya Pradesh, Telangana — are precisely the large agrarian states where that formal-sector shortfall has been most persistent. The concentration of large

gaps in a contiguous agrarian belt, rather than a random scatter across the map, is itself evidence against a purely idiosyncratic or measurement-driven explanation for the state-level pattern.

3.7 Household Credit Portfolios

Moving from the loan level to the household level, Table 7 classifies every sample household into one of four mutually exclusive categories based on whether it reports formal loans, informal loans, both, or neither. Of the 116,461 sample households, 43,640 (37.5% unweighted, 58.9% weighted) report no outstanding loans at all. Among the 72,821 borrowing households, the large majority rely on a single market: 39,930 households (34.3% of all households) borrow only from formal sources, and 21,477 (18.4%) only from informal sources. A smaller but economically important group of 11,414 households (9.8% of all households, 15.7% of borrowing households) access both markets simultaneously.

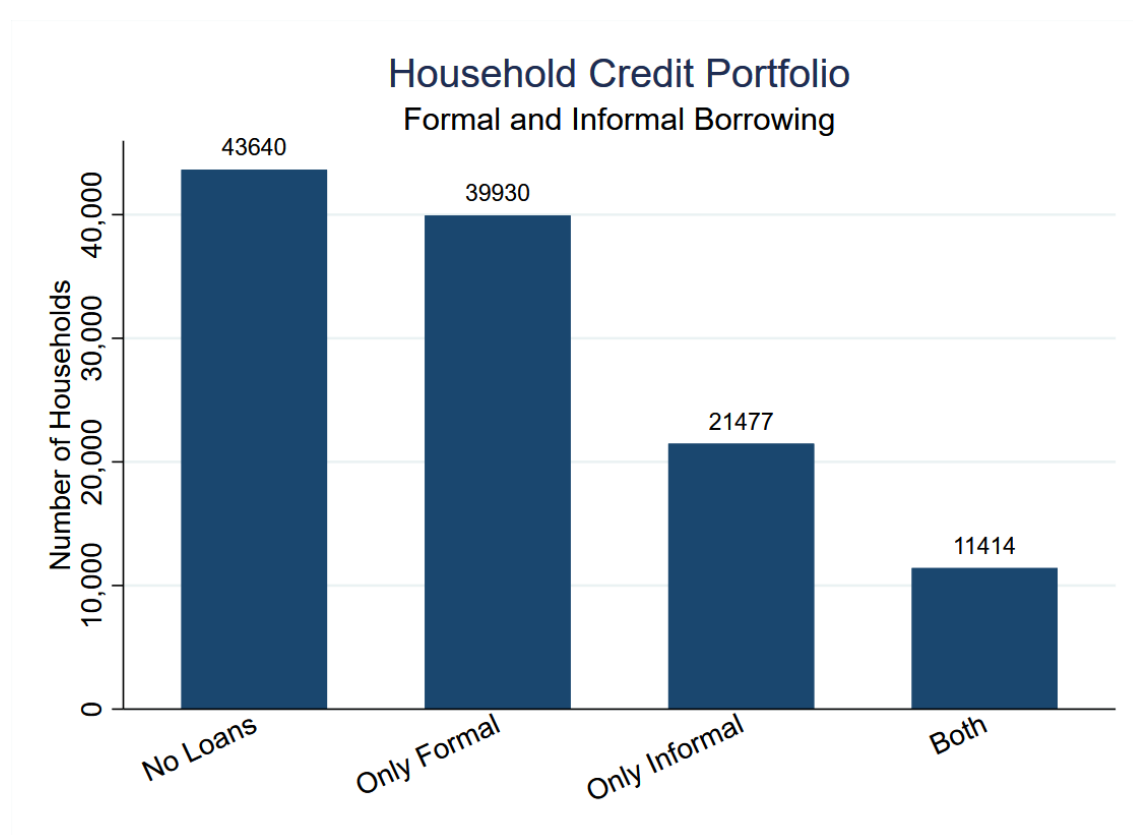


Figure 4. Distribution of households by credit portfolio type

The 9.8% of households holding both formal and informal debt simultaneously are of particular interest because their behaviour is inconsistent with a simple story in which informal credit is used only by those wholly excluded from the formal sector. Rather, this pattern fits the "portfolios of the poor" view of household finance, in which low- and middle-income households routinely combine multiple, imperfectly substitutable credit and savings instruments to manage irregular income and lumpy needs, rather than relying on a single source (Rutherford, 2000; Karlan & Morduch, 2010). Formal and informal credit appear to function as complements for this group: formal credit for larger, longer-term or collateralised needs,

informal credit for the flexible, fast-disbursing top-up rather than as substitutes competing for the same borrower.

3.8 Demographic Profile of Dual-Access Households

Table 8 cross-tabulates portfolio type against rural/urban location. Dual-access ("Both") households are disproportionately rural: 71.4% of "Both" households are rural, compared with 54.7% of Only-Formal and 67.8% of Only-Informal households ($\chi^2(2) \approx 1.6 \times 10^3$, $p < 0.001$). This is consistent with rural households facing thinner, more fragmented credit markets that push them toward combining formal and informal sources rather than relying on either alone.

Table 8. Portfolio type by rural/urban location

Portfolio type	Urban	Rural	Total
Only Formal	18,106 (45.3%)	21,824 (54.7%)	39,930
Only Informal	6,919 (32.2%)	14,558 (67.8%)	21,477
Both	3,261 (28.6%)	8,153 (71.4%)	11,414
Total	28,286 (38.8%)	44,535 (61.2%)	72,821

Table 9 shows the same breakdown by social group. Dual-access households are notably under-represented among ST households (8.8% of "Both" vs. 13.0% of all borrowing households) and over-represented among OBC households (48.5% vs. 41.8%) ($\chi^2(6) \approx 1.5 \times 10^3$, $p < 0.001$). Combined with Table 4's finding that ST households face one of the widest raw interest-rate gaps, this suggests ST households are somewhat less likely to reach dual-market access in the first place, even though those who do face a large gap when they get there a pattern examined further in the household-level regressions below.

Table 9. Portfolio type by social group

Portfolio type	ST	SC	OBC	Other	Total
Only Formal	5,187 (13.0%)	6,088 (15.2%)	15,193 (38.1%)	13,462 (33.7%)	39,930
Only Informal	3,247 (15.1%)	4,016 (18.7%)	9,696 (45.2%)	4,518 (21.0%)	21,477
Both	1,004 (8.8%)	1,994 (17.5%)	5,539 (48.5%)	2,877 (25.2%)	11,414
Total	9,438 (13.0%)	12,098 (16.6%)	30,428 (41.8%)	20,857 (28.6%)	72,821

Education of the household head follows a clear gradient (Table 10): 39.0% of dual-access households have a head with no/below-primary education, compared with just 28.5% of Only-Formal households, while only 7.4% of dual-access households have a graduate head, versus 15.5% for Only-Formal. Less-educated households look more like Only-Informal households in their education profile even when they do reach the formal sector, consistent with education proxying for the financial literacy or documentation needed to access formal credit on more favourable, single-market terms.

Table 10. Portfolio type by education of household head

Portfolio type	No/Below Primary	Primary/ Middle	Secondary/ H.Sec.	Graduate+	Total
Only Formal	11,382 (28.5%)	11,875 (29.7%)	10,469 (26.2%)	6,202 (15.5%)	39,930
Only Informal	8,955 (41.7%)	6,750 (31.4%)	4,283 (19.9%)	1,489 (6.9%)	21,477
Both	4,454 (39.0%)	3,719 (32.6%)	2,394 (21.0%)	847 (7.4%)	11,414
Total	24,791 (34.0%)	22,344 (30.7%)	17,146 (23.6%)	8,538 (11.7%)	72,821

Finally, dual-access households own more land on average (2.01 acres) than either Only-Formal (1.60 acres) or Only-Informal (0.92 acres) households. Owning more land plausibly gives a household both the collateral to access formal credit and, simultaneously, the scale of agricultural operation that drives the informal borrowing documented in Table 5. landed households may be large enough to need financing from both markets at once, rather than substituting one for the other.

Table 11. Land owned by portfolio type (acres)

Portfolio type	N	Mean (acres)	SD	Median
Only Formal	34,903	1.60	3.66	0.24
Only Informal	18,088	0.92	1.86	0.16
Both	10,171	2.01	3.69	0.70
Total	63,162	1.47	3.28	0.27

3.9 Determinants of the Formal–Informal Gap

We next ask what predicts the loan-level formal-informal interest rate gap, using the regression framework set out in Sections 2.2–2.3. The main regression combines the baseline specification (HH-1, Table 12) with its loan-portfolio-composition extension (HH-5, Table 13); a second regression adds the collateral interaction (HH-3, Table 14); and a third regression tests purpose heterogeneity (HH-4, Table 15). We then report the one district-level regression (D-2, Table 16) and the one state-level regression (S-2, Table 17) for which results are currently available the remaining district and state specifications (D-1, D-3, S-1) require external RBI/DBIE and NITI Aayog data that have not yet been merged, so they have not been estimated. Robustness of the baseline informal coefficient to alternative winsorising/trimming rules, to a missing-indicator sample that keeps all 58,949 rated loans, and to a log-rate specification is summarised at the end of this sub-section

3.9.1 Baseline Loan-Level Gap

Table 12 reports the baseline specification (equation 1). Controlling for loan purpose, tenure, security, and the log amount borrowed, and for household size, log MPCE, education, social group, and land owned, and absorbing district fixed effects, informal loans carry an interest rate 15.7 percentage points higher than formal loans. Larger and secured loans receive a modest rate discount (-0.38 pp per log rupee borrowed and -1.12 pp for being secured, respectively), medium- and long-tenure loans carry a small premium over short-tenure loans, and loans for

medical treatment (+4.14 pp) and housing (+2.28 pp) carry the largest purpose premia relative to the omitted purpose category. SC households pay a modest 0.77 pp more than the "Other" baseline even net of these controls.

Table 12. Baseline loan-level hedonic interest-rate regression

Variable	Coef.	(SE)
Informal	15.731***	(0.523)
ln(Amount borrowed)	-0.378***	(0.101)
Secured	-1.121***	(0.264)
Tenure: Medium (1-3y)	1.244***	(0.320)
Tenure: Long (3y+)	1.670***	(0.342)
Purpose: Medical treatment	4.138***	(0.526)
Purpose: Housing	2.278***	(0.302)
HH size	0.089**	(0.041)
ln(MPCE)	-0.397*	(0.212)
Education (max yrs)	-0.071***	(0.022)
SC dummy	0.767**	(0.301)
Land owned (acres)	-0.029*	(0.016)
Constant	16.585***	(1.853)
N	52,093	
R ² / Within R ²	0.583 / 0.486	
District FE	680 clusters	

*Notes: Dependent variable is the annual interest rate (%). Full purpose (12 categories) and tenure (3 categories) dummies, and ST/OBC coefficients, are estimated but omitted from the table for space; district fixed effects absorbed. Robust standard errors clustered on district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

3.9.2 Loan-Portfolio Composition

Table 13 adds household loan-portfolio composition. Households with a larger institutional share of their overall portfolio pay significantly lower rates even on loans that are themselves informal (-0.95) pp per unit of institutional share, consistent with a reputation or relationship spillover from formal-sector standing. The debt-to-asset ratio and the number of loans held are not significant, offering no evidence here that over-indebtedness or portfolio fragmentation alone raise the rate charged. A prior loan write-off is associated with significantly lower, not higher, rates (-1.56 pp) the opposite of the hypothesised default-history penalty plausibly because write-offs are concentrated among subsidised, waived formal agricultural loans, or reflect reverse causality this pattern warrants closer investigation before being read causally.

Table 13. Loan-portfolio composition and the interest rate

Variable	Coef.	(SE)
Informal	16.388***	(0.547)
InstShare	-0.948**	(0.400)
DebtAssetRatio	0.177	(0.131)
NumLoans	0.109	(0.082)
WriteOff (any)	-1.555***	(0.364)
Land owned (acres)	-0.128***	(0.021)
Constant	15.978***	(1.819)
N	52,092	
R ² / Within R ²	0.568 / 0.468	
District FE	680 clusters	

Notes: Household controls (HH size, ln(MPCE), education, ST/SC/OBC) included but not shown. Robust standard errors clustered on district in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.9.3 Collateral and Land Ownership

Table 14 tests the collateral channel. Being secured lowers a loan's rate by 1.67 pp regardless of lender type; the informal $\times$ secured interaction is small and statistically insignificant (-0.27 pp), indicating no evidence that informal lenders discount collateral differently from formal ones in this specification both segments reward security similarly. Land ownership itself (independent of whether this particular loan is secured) is strongly associated with lower rates (-0.23 pp per acre), consistent with a broader reputation/collateral-capacity channel operating outside the specific secured-loan margin.

Table 14. Role of collateral / land ownership in the rate gap (HH-3)

Variable	Coef.	(SE)
Informal	15.968***	(0.553)
Secured	-1.665***	(0.265)
Informal $\times$ Secured	-0.267	(0.846)
Land owned (acres)	-0.229***	(0.030)
HH size	0.128**	(0.053)
ln(MPCE)	-1.244***	(0.301)
Education (max yrs)	-0.085***	(0.028)
Constant	22.140***	(2.414)
N	52,094	
R ²	0.478	

Notes: Household controls include ST/SC/OBC dummies (not shown); no district fixed effects. Robust standard errors clustered on district in parentheses (681 clusters). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3.9.4 Purpose Heterogeneity

Table 15 fully interacts the informal indicator with loan purpose, absorbing district fixed effects (purpose main effects and household controls are estimated but omitted from the table). Every purpose category shows a large, highly significant informal premium, ranging from 13.7 pp (debt repayment) to 20.0 pp (medical treatment); there is no loan purpose for which formal and informal rates converge. The premium is widest for medical treatment (20.0 pp) and financial investment (18.9 pp) classic emergency or consumption-smoothing needs where the borrower has the least bargaining power and narrowest, though still large, for debt repayment, litigation, and education loans, consistent with the hypothesis that urgency raises the informal mark-up specifically, on top of any purpose-specific difference in the formal-sector rate.

Table 15. Informal gap by loan purpose (HH-4)

Purpose × Informal	Gap (pp)
Medical treatment	20.02***
Financial investment	18.92***
Farm revenue expenditure	17.81***
Farm capex	17.65***
Other	16.93***
Housing	15.70***
Non-farm revenue	15.60***
Non-farm capex	15.07***
Other household expenditure	14.81***
Education	14.69***
Litigation	14.64***
Debt repayment	13.71***
N	52,093
R ² / Within R ²	0.582 / 0.486

Notes: $p < 0.01$ for every listed purpose category. District fixed effects absorbed; robust standard errors clustered on district (680 clusters).

3.9.5 District-Level Evidence

Table 16 reports the second-stage district regression described in Section 2.3. The first-stage specification recovers a district-level informal premium of (17.0 pp), close to the baseline estimate in Table 12, confirming that the district fixed effects used in the second stage summarise genuine cross-district variation in credit cost rather than compositional noise. In the second stage, districts with higher average MPCE and more land owned per household show a significantly lower residual district credit cost, consistent with wealthier, more asset-rich districts having more competitive or accessible formal credit and less reliance on high-cost informal lenders. The ST/SC population share is negative and only marginally significant districts with a larger ST/SC population do not show a higher residual credit cost at the district

level, which cuts against a simple district-level-discrimination story. The rural population share is not significant once MPCE, land, and social composition are controlled for.

Table 16. District fixed effects on district characteristics, second stage (D-2)

District covariate	Coef.	(SE)
Avg. MPCE	-0.00046**	(0.00018)
Avg. land owned	-0.615***	(0.107)
ST/SC population share	-1.433*	(0.758)
Rural share	-1.208	(0.889)
Constant	3.181***	(1.076)
N (districts)	680	
R ²	0.062	

Notes: Dependent variable is the district fixed effect δ_d recovered from a leaner first-stage version of equation (1) (informal indicator plus household controls only, no loan controls). robust (HC) standard errors.

3.9.6 State-Level Evidence: Rural-Urban Heterogeneity

Table 17 re-estimates the baseline specification with state fixed effects in place of district fixed effects and a rural $\times$ informal interaction. Within the same state, urban formal loans are the cheapest baseline; rural loans are 1.71 pp cheaper than urban loans on the formal margin, but this advantage reverses for informal loans: the informal $\times$ rural interaction is positive and marginally significant (+3.22 pp), indicating that rural households face a steeper informal penalty than urban households even after absorbing all state-level regulatory and banking-structure differences. This corroborates the earlier descriptive pattern in Table 2 (rural gap of 8.97 pp versus an urban gap of 5.82 pp) with a regression-based, within-state estimate: rural credit markets are not simply poorer on average but specifically more costly for informal borrowing.

Table 17. Household regression with state fixed effects and rural-urban interaction

Variable	Coef.	(SE)
Informal	14.417***	(1.399)
Rural	-1.706***	(0.428)
Informal × Rural	3.219*	(1.820)
HH size	0.125***	(0.039)
ln(MPCE)	-0.405	(0.302)
Education (max yrs)	-0.120***	(0.027)
ST dummy	0.866	(0.913)
SC dummy	1.665***	(0.363)
OBC dummy	0.310	(0.253)
Land owned (acres)	-0.143***	(0.026)
Constant	15.864***	(2.472)
N	52,094	
R ² / Within R ²	0.509 / 0.467	
State FE	36 clusters	

Notes: Robust standard errors clustered on state in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4. Discussion

Taken together, the descriptive and regression results point to a consistent story. Informal credit is, on average, roughly 70% more expensive than formal credit (Table 2), but this average masks two very different sources of variation. First, which informal lender a household uses matters enormously: professional moneylenders and landlords charge 27-32%, while borrowing from relatives and friends is typically interest-free (Table 3). Second, the size of the gap is strongly tied to what the loan is for, not just who the borrower is: the loan-level regressions confirm that the gap is large and highly significant for every loan purpose (Table 15), and is in fact widest for medical treatment and financial-investment loans classic emergency and consumption-smoothing needs where borrowers have the least bargaining power with agricultural purposes close behind, rather than concentrating narrowly in agriculture as the raw loan-level tabulation alone might suggest (Table 5).

The household-portfolio analysis adds a selection dimension the loan-level regressions alone can't show. Dual-access households are disproportionately rural, land-owning, and headed by less-educated individuals (Tables 8, 10, 11) profiles that plausibly reflect selection into which households end up borrowing in both markets in the first place. The raw social-group gap in Table 4, and the corresponding under-representation of SC/ST households among dual-access borrowers in Table 9, are consistent with an access-based story. The loan-level regressions in this paper do not yet decompose that social-group gap into a pricing versus a selection component the closest recent attempts, using comparable data, are Zeeshan et al. (2025) and Sangwan and Saha (2026), discussed in Section 1 but the district-level second stage in Table 16 at least rules out one simple version of the access story: it finds no relationship between a

district's ST/SC population share and its residual credit cost, so whatever drives the raw caste gap is not just a matter of which districts SC/ST households happen to live in.

These findings also speak to the literature reviewed in Section 1. The information- and enforcement-based theories of informal lending (Aleem, 1990; Hoff & Stiglitz, 1990; Bell, 1990) are broadly consistent with the pattern in Table 3, where the cheapest informal category (relatives/friends) is precisely the one relying least on formal screening or collateral and most on repeated social interaction, while the most expensive categories (moneylenders, landlords) are the ones most plausibly relying on local information rents, interlinkage, or limited local competition. The branch-expansion literature (Burgess & Pande, 2005; Briones et al., 2024) helps explain the geographic concentration of large gaps in Table 6 and Figure 3. The caste-and-credit-access literature (Deshpande, 2011; Kumar, 2016) is consistent with the raw pattern in Tables 4 and 9, and the more recent decompositions by Zeeshan et al. (2025) and Sangwan and Saha (2026), both discussed in Section 1, point in the same direction though testing whether the caste gap survives conditioning on loan and household characteristics, the way the purpose and collateral results in Section 3.9 do, is left for future work. Finally, the directed-agricultural-credit literature (Shah, Rao, & Vijay Shankar, 2007) offers an important caveat for interpreting the agricultural-purpose results: part of the wide agricultural gap in Table 5 is a mechanical consequence of subsidised formal agricultural lending rather than solely evidence of a more predatory informal agricultural credit market though the loan-level result in Table 15, which holds loan, household, and district characteristics fixed, shows farm capex and farm-revenue loans still carrying a 17.6-17.8 percentage-point informal premium, nearly as large as the single widest category (medical treatment, 20.0 pp). That suggests a substantively real effect survives this caveat, even if agricultural loans are no longer the single largest purpose-specific premium once other covariates are controlled.

5. Conclusion

This paper has documented the size, distribution, and correlates of the gap between formal and informal interest rates paid by Indian households, using loan-level data from AIDIS 2019. The headline finding that informal credit costs roughly 70% more than formal credit on average sits atop substantial heterogeneity: by lender type, where kinship-based lending is free and moneylender/landlord lending is very expensive; by loan purpose, where every purpose carries a large and significant informal premium but medical treatment and financial-investment loans carry the widest; by social group, where SC and ST households face the widest raw gaps (Table 4); and by geography, where the gap concentrates in a contiguous belt of large agrarian states and is systematically steeper in rural than urban credit markets within the same state. At the household level, roughly one in ten households holds both formal and informal debt simultaneously, and this dual-access group is disproportionately rural, less educated, and land-owning.

The paper's central empirical contribution is to show that the formal-informal gap survives a wide range of loan- and household-level controls rather than being explained away by them. The main regression puts the baseline gap at 15.7 percentage points (Table 12) and shows, in its portfolio-composition extension, that the gap is not driven by household over-indebtedness

or portfolio fragmentation (Table 13); a second regression shows the gap is not meaningfully narrowed by collateral (Table 14); and a third regression shows the gap is large and significant for every loan purpose (Table 15).

The gap is also significantly steeper for rural than urban households within the same state (Table 17), even after absorbing state-level regulatory and banking-structure differences. At the district level, wealthier and more land-rich districts have a significantly lower residual credit cost, while a district's ST/SC population share and rural share are not significant predictors of that residual cost (Table 16), suggesting that geography and household wealth, rather than local social composition, are the main district-level correlates of the gap.

This distinction matters for policy: closing the gap is likely to require interventions targeted at loan purpose and rural credit markets specifically, and at district-level banking and asset conditions, rather than assuming that improving household collateral or reducing portfolio risk alone will close it. Decomposing the raw social-group gap documented in Table 4 into pricing and selection components remains an open question for future work.

References

- Aleem, I. (1990). Imperfect information, screening, and the costs of informal lending: A study of a rural credit market in Pakistan. *World Bank Economic Review*, 4(3), 329–349.
- Banerjee, A. V., & Duflo, E. (2010). Giving credit where it is due. *Journal of Economic Perspectives*, 24(3), 61–80.
- Basu, K. (1997). *Analytical development economics: The less developed economy revisited*. MIT Press.
- Bell, C. (1990). Interactions between institutional and informal credit agencies in rural India. *World Bank Economic Review*, 4(3), 297–327.
- Briones, L., Chiseliov, V., Gushchin, E., Guttesen, J., & Narayanaswamy, K. (2024). Financial accessibility and interest rates in the informal credit market in India (Working paper). Barcelona School of Economics. <https://doi.org/10.21203/rs.3.rs-3855411/v1>
- Burgess, R., & Pande, R. (2005). Do rural banks matter? Evidence from the Indian social banking experiment. *American Economic Review*, 95(3), 780–795.
- Deshpande, A. (2011). *The grammar of caste: Economic discrimination in contemporary India*. Oxford University Press.
- Ghatak, M., & Guinnane, T. W. (1999). The economics of lending with joint liability: Theory and practice. *Journal of Development Economics*, 60(1), 195–228.
- Hoff, K., & Stiglitz, J. E. (1990). Imperfect information and rural credit markets: Puzzles and policy perspectives. *World Bank Economic Review*, 4(3), 235–250.
- Hoffmann, V., Rao, V., Surendra, V., & Datta, U. (2021). Relief from usury: Impact of a self-help group lending program in rural India. *Journal of Development Economics*, 148, 102567.
- Karlan, D., & Morduch, J. (2010). Access to finance. In D. Rodrik & M. Rosenzweig (Eds.), *Handbook of Development Economics* (Vol. 5, pp. 4703–4784). Elsevier.
- Kumar, S. M. (2016). Why does caste still influence access to agricultural credit? (WIDER Working Paper 2016/86). UNU-WIDER.
- National Sample Survey Office. (2021). *All India Debt and Investment Survey, 2019 (NSS 77th Round)*. Ministry of Statistics and Programme Implementation, Government of India.
- Rutherford, S. (2000). *The poor and their money*. Oxford University Press.
- Sangwan, N., & Saha, B. (2026). Caste and credit: Discrimination in India's credit sector. *Progress in Development Studies*. Advance online publication. <https://doi.org/10.1177/14649934251411694>
- Shah, M., Rao, R., & Vijay Shankar, P. S. (2007). Rural credit in 20th century India: Overview of history and perspectives. *Economic and Political Weekly*, 42(15), 1351–1364.
- Swaminathan, M. (1991). Segmentation, collateral undervaluation, and the rate of interest in agrarian credit markets: Some evidence from South India. *Cambridge Journal of Economics*, 15(2), 161–178.